

Supplementary information

Reading-frame constraints and retained RG content in NR4A3 fusion models of extraskeletal myxoid chondrosarcoma

These tables retain the submitted Research Square v3 scientific content. References use the numbering in the main article.

Supplementary tables

Three supplementary tables are proposed. S1 and S2 carry content moved from the main text without change. S3 is new to this revision and is assembled from one committed artifact; it introduces no construct that the artifact does not already hold.

Supplementary Table S1. Reference gene models, from the UniProt and Ensembl records retrieved into the input cache [8]. Transcript accessions are the ones the cache holds; the cache records its retrieval as 2026-08-12. No assembly, release or version suffix is given, because none is recorded.

gene	transcript	protein	transcript / coding exons	Ensembl matches UniProt
EWSR1	ENST00000397938	656 aa	17 / 17	yes
TAF15	ENST00000605844	592 aa	16 / 16	yes
FUS	ENST00000254108	526 aa	15 / 15	yes
NR4A3	ENST00000395097	626 aa	8 / 6, exons 1-2 non-coding	yes
TCF12	ENST00000333725	706 aa	21 / 19	no; UniProt Q99081 is 682 aa

Supplementary Table S2. Wild-type controls and their predictions. Full-length sequences are in the artifact under wild_type_controls.

control	role	prediction
GFP-EWSR1, full length	fast-recruitment anchor, already held by any laboratory running the assay	rapid recruitment, as published. Failure to reproduce it makes nothing else in the run interpretable
GFP-TAF15, full length	wild-type anchor for the TAF15::NR4A3 arm	rapid recruitment, as TAF15 carries its own C-terminal RGG region. Not previously reported in this assay, so a prediction rather than a reproduction

control	role	prediction
GFP-NR4A3, full length	partner-alone control, the EMC analogue of reference 1's GFP-FLI1 control	no accumulation. Recruitment of NR4A3 alone would show that the partner reaches breaks on its own; it would not by itself remove a FET contribution in the fusion, which a partner-alone control cannot isolate
GFP-TCF12, full length	non-FET partner-alone control	no accumulation, TCF12 being non-FET by section 3.5. It is not a TCF12::NR4A3 construct and does not test P5, which stays untested without one

Supplementary Table S3. Per-junction assembly coordinates for each reported junction with a sourced transcript-level breakpoint. Every cell is a verbatim copy of a recorded leaf of emc-fet-construct-designs.json (sha256 726aae02ae38b41c34d4398363e3581cfc9e4602d0fe9d65906a4fba79c2048b), a fixed column label, or UNRESOLVED. No value in this table was derived, inferred, rounded or interpolated, and no new construct is proposed by it.

Field	EWSR1_NR4A3_type 1	EWSR1_NR4A3_type 2	EWSR1_NR4A3_type 5	TAF15_NR4A3
Junction (label)	EWSR1::NR4A3 type 1 — EWSR1 exon 12 :: NR4A3 exon 3	EWSR1::NR4A3 type 2 — EWSR1 exon 7 :: NR4A3 exon 2	EWSR1::NR4A3 type 5 — EWSR1 exon 13 :: NR4A3 exon 3	TAF15::NR4A3 — TAF15 exon 6 :: NR4A3 exon 3
Reported rank — LEGACY SOURCE DESCRIPTION, not a counted frequency (see note)	the commonest reported EWSR1::NR4A3 transcript type	the second reported EWSR1::NR4A3 transcript type	a minority reported type	the only reported TAF15::NR4A3 coding junction
5' transcript	ENST00000397938	ENST00000397938	ENST00000397938	ENST00000605844
5' last exon retained (transcript rank)	12	7	13	6
3' transcript	ENST00000395097	ENST00000395097	ENST00000395097	ENST00000395097
3' first exon retained (transcript rank)	3	2	3	3
CUT — 5' cDNA nt retained	1363	862	1486	570
CUT — 5' coding nt retained	1294	793	1417	484
5' UTR length (nt)	69	69	69	86
CUT — 3' cDNA resume offset (0-based)	697	523	697	697
3' UTR nt read through	2	176	2	2
Assembled cDNA length (nt)	6270	5943	6393	5477
ORF length (nt)	3174	2847	3297	2364

Field	EWSR1_NR4A3_type 1	EWSR1_NR4A3_type 2	EWSR1_NR4A3_type 5	TAF15_NR4A3
ORF length (aa)	1058	949	1099	788
5' residues fully encoded	431	264	472	161
Codon split across junction	yes	yes	yes	yes
Seam residue index (1-based)	432	265	473	162
Junction context (aa)	TAKAAVEWFD DMP CVQAQYS	SQSSSYGQQ KPTA EEGSPA	GRGMPPLRG DMP CVQAQYS	QRENYSHHTQ DMP CVQAQYS
Junction context (nt)	AATGGTTTGATG AT ATGCCCTGCG	ACGGGCAGCAGA A GCCCACTGCGG	CACTCCGTGGAG A TATGCCCTGCG	ACCACACACAAG A TATGCCCTGCG
Extra junction-encoded residues	1	59	1	1
In frame (self-check)	yes	yes	yes	yes
5' start matches partner (self-check)	yes	yes	yes	yes
3' C-terminus intact (self-check)	yes	yes	yes	yes
CUT — genomic breakpoint coordinates	UNRESOLVED	UNRESOLVED	UNRESOLVED	UNRESOLVED

Supplementary Table S3, continued. Reported fusions with no sourced transcript-level junction.

Fusion	Every assembly field	Recorded status
FUS::NR4A3	UNRESOLVED	NO transcript-level junction sourced in this repo's literature cache
TCF12::NR4A3	UNRESOLVED	GENOMIC breakpoint only — reported as TCF12 intron 5, not as an mRNA exon junction, and TCF12 has several alternatively-spliced isoforms

Note on the "Reported rank" row. Its four cells are verbatim leaves of the artifact and are retained unchanged, with their provenance, but they are a legacy source description of how each junction was characterised, not a frequency counted in this work. Table 1 carries the corrected frequency evidence, with each count attached to the series it comes from; where the two differ, Table 1 governs. No artifact value was modified to add this label.

Genomic breakpoint coordinates are UNRESOLVED for all four junctions because the input carries transcript and cDNA coordinates only; they are not back-derived from exon ranks. FUS::NR4A3 and TCF12::NR4A3 appear in no column because the input records no transcript-level junction for either — TCF12 is reported at genomic

resolution only, in intron 5, and TCF12 has several alternatively spliced isoforms. That is a limit of the sourcing and not evidence about the fusions.

References

1. Gracilla DE, Menon S, Breese MR, Lin YP, Dela Cruz FS, Feinberg TY, et al. FET Fusion Oncoproteins Disrupt Physiologic DNA Repair and Create a Targetable Opportunity for ATR Inhibitor Therapy. *Cancer Research* 2026;86:2660-2677. PMID 41811428. PMC13223543. doi 10.1158/0008-5472.can-25-2166.
2. Remiszewski P, Falkowski S, Szumera-Ciećkiewicz A, Spałek MJ, Rutkowski P, Czarnecka AM. From pathogenesis to the patient's bedside: a comprehensive review of extraskeletal myxoid chondrosarcoma. *J Cancer Res Clin Oncol* 2025;151(11):283. PMID 41055792. PMC12504171. doi 10.1007/s00432-025-06316-5.
3. Brenca M, Stacchiotti S, Fassetta K, Sbaraglia M, Janjusevic M, Racanelli D, et al. NR4A3 fusion proteins trigger an axon guidance switch that marks the difference between EWSR1 and TAF15 translocated extraskeletal myxoid chondrosarcomas. *J Pathol* 2019;249(1):90-101. PMID 31020999. PMC6766969. doi 10.1002/path.5284.
4. Nishio J, Iwasaki H, Nabeshima K, Naito M. Cytogenetics and molecular genetics of myxoid soft-tissue sarcomas. *Genet Res Int* 2011;2011:497148. PMID 22567356. PMC3335514. doi 10.4061/2011/497148.
Source of the verbatim type 1 and type 2 exon-level definitions and of the TAF15 exclusivity statement quoted in section 3.1.
5. Cerrone M, Cantile M, Collina F, Marra L, Liguori G, Franco R, et al. Molecular strategies for detecting chromosomal translocations in soft tissue tumors (review). *Int J Mol Med* 2014;33(6):1379-1391. PMID 24714847. PMC4055444. doi 10.3892/ijmm.2014.1726. Source of the type 5 definition, the second TAF15 exclusivity statement, and the TCF12 genomic-only intron 5 breakpoint quoted in section 3.1.
6. Agaram NP, Zhang L, Sung YS, Singer S, Antonescu CR. Extraskeletal myxoid chondrosarcoma with non-EWSR1-NR4A3 variant fusions correlate with rhabdoid phenotype and high-grade morphology. *Hum Pathol* 2014;45(5):1084-1091. PMID 24746215. PMC4015728. doi 10.1016/j.humpath.2014.01.007.
7. Panagopoulos I, Mertens F, Isaksson M, Domanski HA, Brosjö O, Heim S, et al. Molecular genetic characterization of the EWS/CHN and RBP56/CHN fusion genes in extraskeletal myxoid chondrosarcoma. *Genes Chromosomes Cancer* 2002;35(4):340-352. PMID 12378528. doi 10.1002/gcc.10127. Counted series. The quotation retained in the frame-and-composition artifact gives counts without a denominator: type 1 in 10 tumours, the most frequent transcript, and type 5 in two cases, the second most common. The denominator of 15 EWS/NR4A3 cases is recorded in this repository's editorial note on this reference, not in that quotation.

8. UniProt and Ensembl reference records for EWSR1 (ENST00000397938), NR4A3 (ENST00000395097), TAF15, FUS and TCF12 (UniProt Q99081), as retrieved into the input cache in section 7.
9. Okamoto S, Hisaoka M, Ishida T, Imamura T, Kanda H, Shimajiri S, Hashimoto H. Extraskeletal myxoid chondrosarcoma: a clinicopathologic, immunohistochemical, and molecular analysis of 18 cases. *Hum Pathol* 2001;32(10):1116-1124. PMID 11679947. doi 10.1053/hupa.2001.28226. Counted series: fusion transcripts detected in 15 of 18 cases, EWS-CHN type 1 in 11, type 2 in 1 and TAF2N-CHN in 3.
10. Sjögren H, Meis-Kindblom JM, Orndal C, Bergh P, Ptaszynski K, Aman P, Kindblom LG, Stenman G. Studies on the molecular pathogenesis of extraskeletal myxoid chondrosarcoma - cytogenetic, molecular genetic, and cDNA microarray analyses. *Am J Pathol* 2003;162(3):781-792. PMID 12598313. PMC1868116. Counted series, by partner gene rather than by EWSR1 subtype: EWS-TEC five cases, TAF2N-TEC four, TCF12-TEC one, of ten tumours.